NAME: ____________________ SECTION DAY/TIME: ____________________________
GSI: ___________________________ LAB PARTNERS: ____________________________

Lab 6: Introduction to oscilloscope and time dependent circuits

Introduction

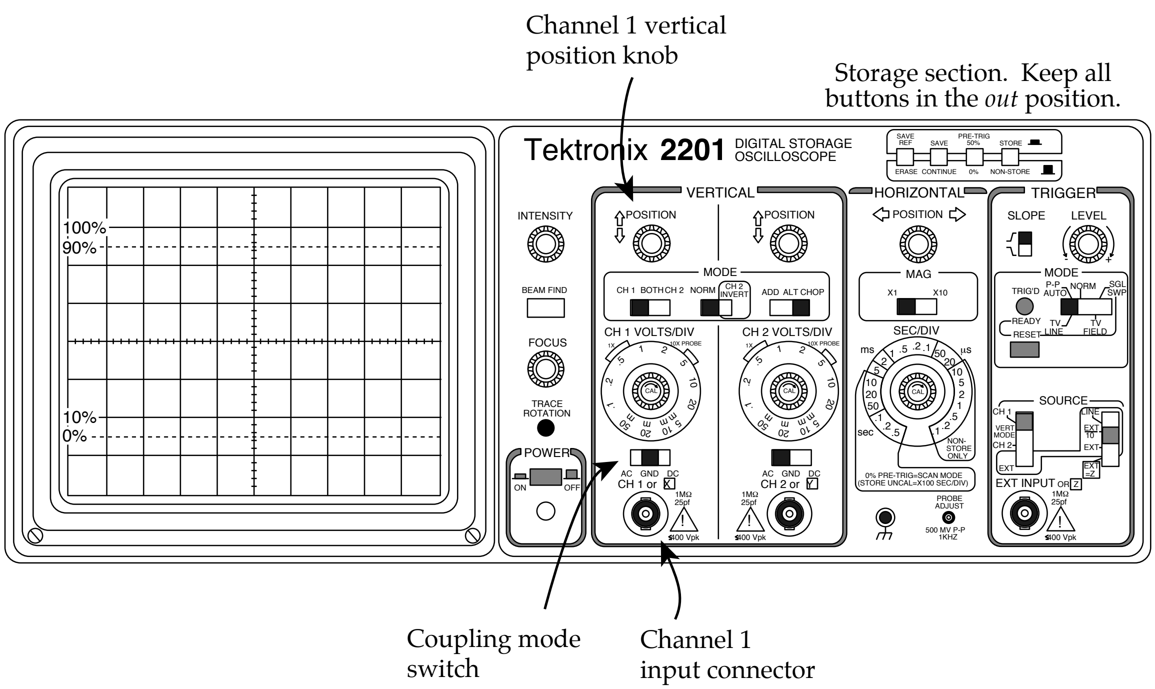
In this lab, you'll learn the basics of how to use an oscilloscope. Then you'll investigate time dependent circuits. When dealing with capacitors and inductors in DC circuits, it's easy to get lost in mathematics, without understanding what's going on conceptually. These questions and lab activities are designed to help you develop an understanding of these circuits, allowing you to address conceptual questions without plugging through unnecessary math. You'll also see what these circuit components look like in real life.

Part I of this experiment, on the basics of the oscilloscope, should take approximately 30 minutes. The rest of your time in lab should be spent working on Part II, on the time dependent RC and LR circuits. (Don't worry if you aren't fully comfortable with the scope by the end of Part I. You'll get more practice in Part II.)

Part I: Oscilloscope Basics

Activity 1: Reset the oscilloscope

- ♦ Turn on the oscilloscope, and disconnect any probes plugged into the "channel 1" (CH 1) input connector.
- ♦ Set all the levers and buttons as indicated here, if they're not already.



- Set the CH 1 coupling mode switch to "ground" (GND).
- Turn down the INTENSITY knob, if necessary, to avoid burning out the screen. The sweeping dot should be clear but not too bright.

Since channel 1 is now "grounded" to zero volts, the oscilloscope should read zero on the vertical axis (using the coordinate axes centered on the screen). If it doesn't. . .

- Adjust the channel 1 vertical POSITION knob so that the oscilloscope reads 0 volts.

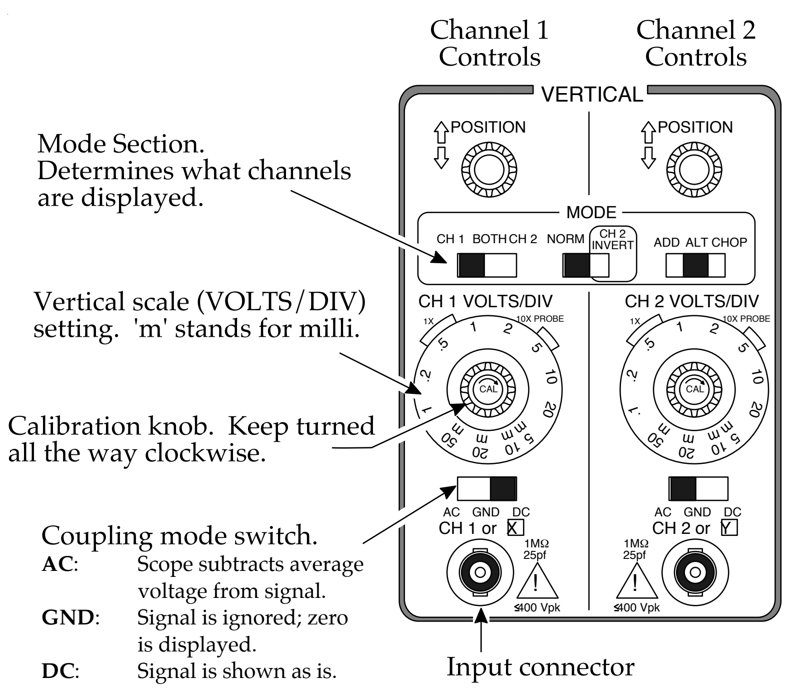
What the oscilloscope does

The oscilloscope graphs voltage vs. time, by sweeping an electron beam across the phosphor screen. Wherever the beam hits the screen, it glows green. For most measurements, the beam sweeps rightward at a constant rate. As you can see, when the beam gets to the right-hand side of the screen, it jumps back to the left-hand side. In this way, the horizontal axis shows time.

When a probe is plugged into the CH 1 input connector, the vertical axis shows the potential difference—i.e., the voltage—between the two wires coming out of that probe. If you're interested, ask your GSI what's going on inside the oscilloscope to deflect the electron beam up or down. Better yet, see if you can figure it out! Hint: Parallel-plate capacitor.

Activity 2: Measuring DC voltages, and using the VOLTS/DIV setting

The point of this brief activity is to practice measuring a voltage with the oscilloscope, and to get a feel for what the VOLTS/DIV control does.



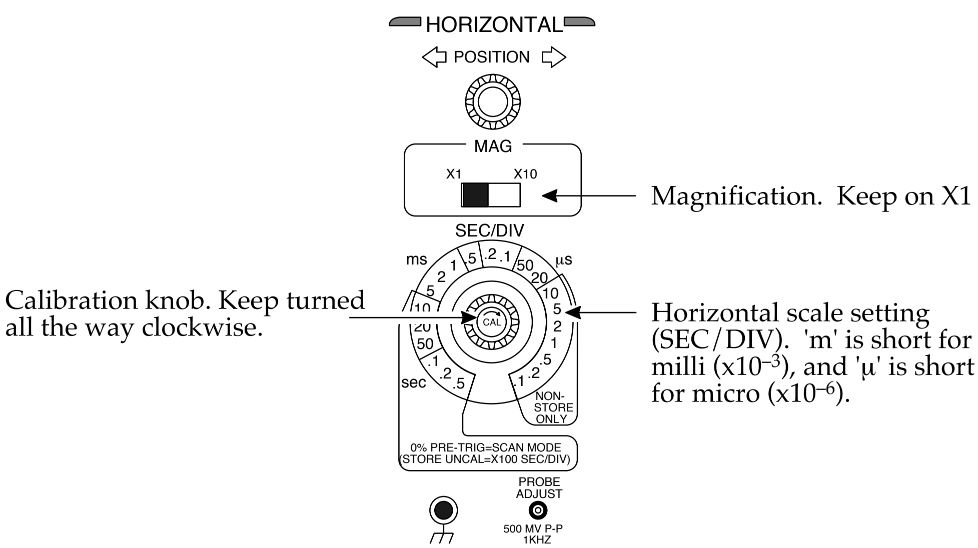
- Set the CH 1 VOLTS/DIV to 2, by aligning the "2" next to the "1X" bracket.
- Set the CH 1 coupling mode switch to DC.
- Now use the oscilloscope to measure the voltage across a 1.5-volt battery.

Make sure you understand what the VOLTS/DIV setting is doing. Students often err in thinking in terms of DIV/VOLT instead of VOLT/DIV.

1. To get a more precise reading of the battery's voltage, should you turn the VOLTS/DIV knob clockwise or counterclockwise? Why? Try it, to get a feel for how much precision can be gained.

Activity 3: Measuring AC voltages, and the SEC/DIV setting

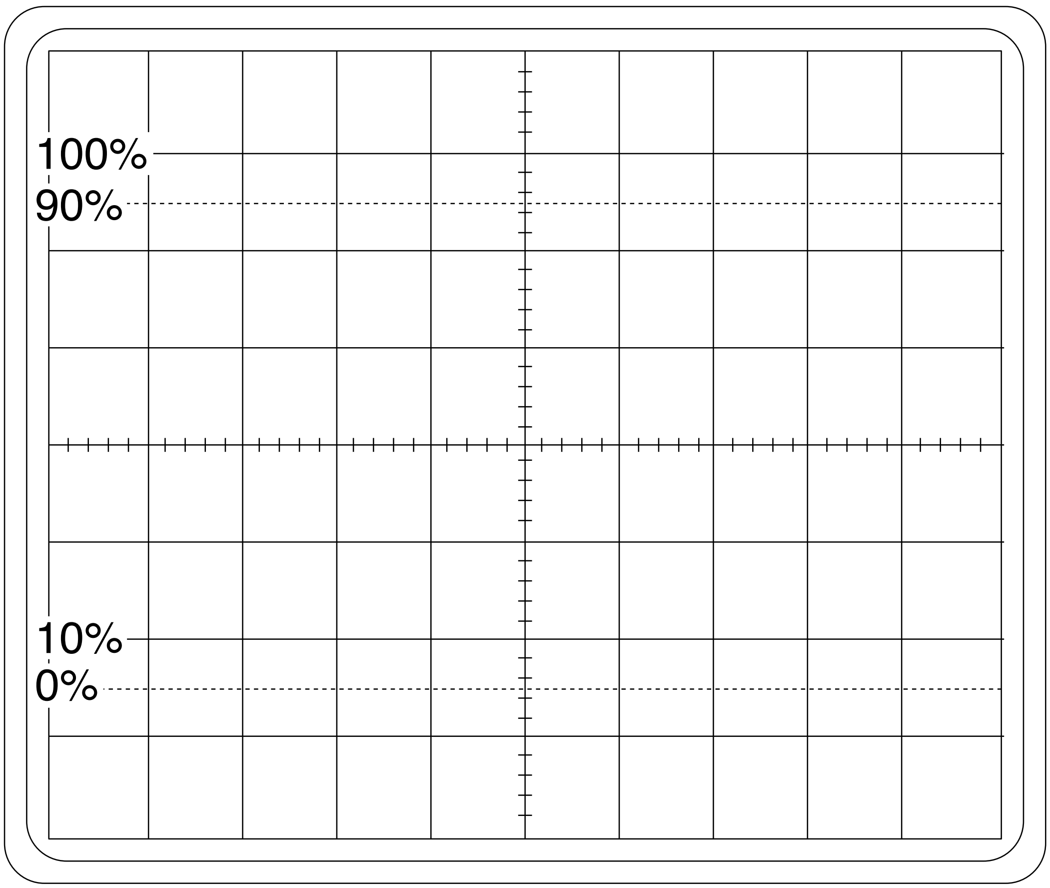
Now you'll practice using an AC power supply, and you'll figure out what the SEC/DIV knob does. The "AC" means "Alternating Current"—that is, the voltage put out by the power supply oscillates with a frequency that you set.



- Set SEC/DIV to 0.5 milliseconds.
- Set the CH 1 VOLTS/DIV to 5.

- Turn on the AC signal generator. Set it to sinusoidal wave, 1000 Hz (i.e., 1.0 kHz). (Note: make sure the sweep width knob is all the way to the left, so it clicks.) But don't connect the AC signal generator to the oscilloscope, until answering this question. . .

2. When you use the oscilloscope to measure the voltage produced by this AC signal generator, what will the screen look like? Sketch your detailed prediction on the next page, paying attention to the amplitude and "wavelength."



- Now hook up the AC power supply to the oscilloscope.

If your prediction was wrong, see if you can figure out what's going on, or get help from your GSI. Sketch the actual screen display in a different color.

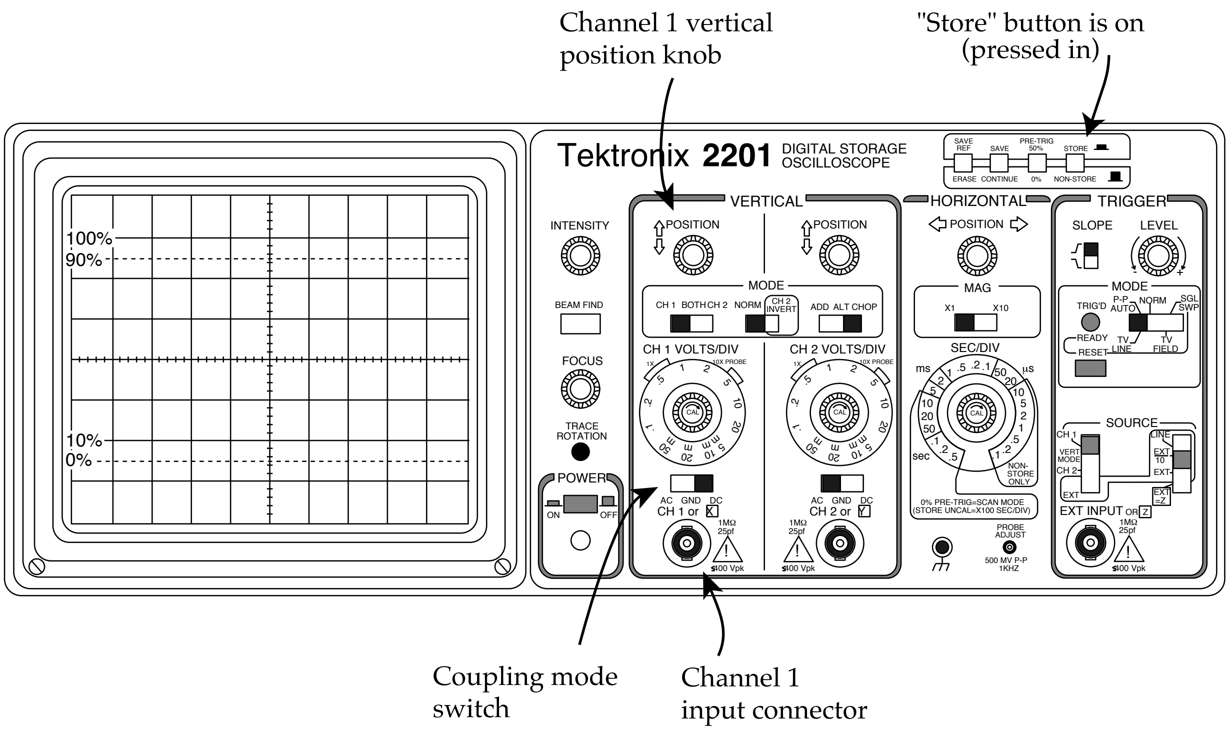
3. To get a more precise measurement of the period of the oscillating voltage, should you turn the SEC/DIV knob clockwise or counterclockwise? Try it.

Part II: Time dependent RC and LR circuits

NOTE: The remainder of the lab is probably too long for the time you have left; your GSI will direct you to which parts of the lab you must complete. Make sure you understand at least questions 1 through 4 before you leave.

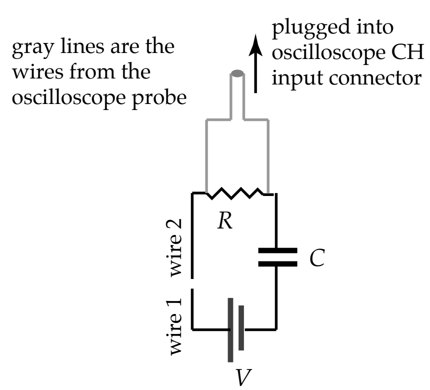
Technical stuff

Adjust the oscilloscope as shown here.



♦ Set SEC/DIV to .5 seconds.
♦ Set CH 1 VOLTS/DIV to .5 volts.

In all of the experiments, you'll build a simple circuit, and then use the oscilloscope probe to measure the voltage (potential difference) across a circuit element as a function of time. For instance, this set-up shows how you'd measure the voltage across the resistor in an RC circuit. Notice that the circuit starts out "open"; current cannot yet flow around it. You'll "close" the circuit by touching wire 1 to wire 2.

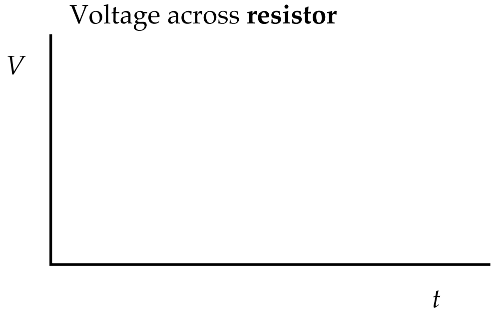


1. Consider a simple RC circuit, with the battery, resistor, and capacitor hooked up in series. Suppose you want to use the oscilloscope to measure the <u>current</u> through this circuit as a function of time. *How can you do it?* (Remember, the oscilloscope can only be used to graph the *voltage* across one or more circuit elements.) We want the graph to have the right general shape; but it need not be scaled properly. In other words, it can be "too tall" or "too short," as long as it has the right shape.

2. For this RC circuit, how can you get the oscilloscope to measure the *charge* on the capacitor as a function of time?

3. Suppose the capacitor is initially uncharged, and the circuit is closed at time t = 0. As your prediction, draw a rough sketch of the voltage across the *resistor* as a function of time, and <u>explain</u> your reasoning.

RC CIRCIT WITH BATTERY
Voltage across **resistor**



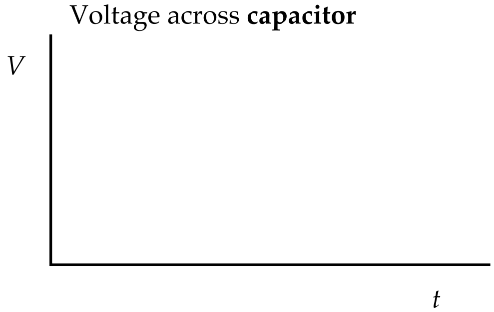
Now do the experiment, using a 1.5-volt battery, a microfarad (10^{-6} F) capacitor, and a megaohm (1 MΩ = 10^6 Ω) resistor. A 1 MΩ resistor is marked with color bands that are **brown, black, and <u>green</u>.** (There are other resistors that have brown, black and brown bands that we will use later in the lab. Don't use this now, since it is only a 100 Ω resistor.)

If the actual result differs from your prediction, sketch it on the graph as a dashed line, and explain what's going on below. **Before closing the circuit, make sure the capacitor is discharged, as demonstrated by your GSI. Each time you redo the experiment, discharge the capacitor again, so that it starts out with zero charge.**

TECHNICAL NOTE: because the oscilloscope has a 1 MΩ resistor at its input, which is in parallel with the 1 MΩ resistor in your circuit, the equivalent resistance of your circuit with the scope attached is (1/2)MΩ. Hence the time constant for your circuit will be half of what you were expecting. We are not concerned with this for the experiment.

4. Same as question 3, but now consider the voltage across the *capacitor* as a function of time. Graph and explain your prediction.

RC CIRCIT WITH BATTERY
Voltage across **capacitor**



Now run the experiment. Re-graph and re-explain, if the results differ from your prediction.

How is the voltage across the capacitor related to the voltage across the resistor as a function of time? Explain.

5. Suppose you place two 1-μF capacitors in series. Is the total capacitance now 2 μF or 0.5 μF? Don't just plug in a formulas; explain your answer conceptually, using diagrams and words. Hint: remember that $Q = C\,\Delta V$.

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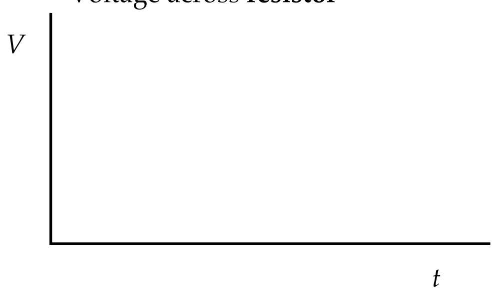
6. Now consider an LR circuit, in which a battery, a resistor, and an inductor are hooked up in series. As you saw in question 1 above, graphing the voltage vs. time across the *resistor* tells you the current through the circuit as a function of time. That's because the voltage across the resistor is proportional to the current ($V = iR$). If the circuit is closed at time $t = 0$, what does the voltage vs. time graph across the resistor look like? Sketch and explain your prediction.

LR CIRCIT WITH BATTERY
Voltage across **resistor**



To do the experiment, replace the capacitors with a 4 H inductor, and replace the mega-ohm resistor with a 100 Ω resistor. The 100 Ω resistor is marked with bands that are **brown, black, and brown**.

Remember to put the oscilloscope probe across the resistor, not across the inductor. For best results, you may want to change the SEC/DIV setting to .1 seconds or even 50 milliseconds (ms). Also, lower the VOLTS/DIV setting to 50 millivolts. Does the graph come out as you expected?

7. Your inductor has an inductance of 4 H and a resistance of about 330 Ω. As you saw in question 6, the circuit's current eventually "settles" to some final value. If you replaced this inductor with a 330 Ω resistor, how would the graph of current vs. time differ from the one in question 6? Specifically,

(a) would the current shoot up to its final value more abruptly or less abruptly than it did in question 8? Explain.

(b) Would the current settle at the same final value as it did in question 8? Or would it settle at a higher or lower final value? Explain.

You need not test your predictions.